JIYI WANG

Georgia Institute of Technology. jwang3617@gatech.edu, geertswon@gmail.com

RESEARCH FOCUS & TECHNICAL SKILLS

Embodied AI / Robot Learning: VLA models, imitation learning, behavior cloning, actor–critic RL, human-in-the-loop RL, inverse RL, world models, dexterous manipulation, humanoid control.

Tools: Python, PyTorch, JAX, C, C++, NVIDIA GRooT, QwenOFT, $\pi_{0.5}$, StarVLA, Isaac Sim/PhysX, MuJoCo, Codex/Claude CLI, Unitree G1 humanoid deployment and control.

EDUCATION

Georgia Institute of Technology Ph.D. in Computer Science and Engineering *May 2026–Present* GPA: 4.0/4.0
Peking University B.S. in Chemistry *Sept 2020–July 2024* GPA: 3.70/4.0

SELECTED PUBLICATIONS

Leveraging Dale's Principle as an Inductive Bias in Recurrent Neural Networks. NeurIPS 2026. *First author.*

Learning Reusable Motor Motifs for Continuous Animal Behavior Modeling. NeurIPS 2026. *First author.*

Inverse Reinforcement Learning with Switching Rewards and History Dependency for Characterizing Animal Behaviors. ICML 2025. *Co-author.*

RESEARCH EXPERIENCE [WEBSITE]

Research Assistant, Georgia Institute of Technology **Atlanta, GA**
Human-in-the-Loop VLA Learning for Dexterous Manipulation, supervised by Dr. Ye Zhao *May 2026–Present*

- Developed learning and control pipelines for clinical robotic manipulation tasks, including high-precision pipetting.
- Fine-tuning VLA (GRooT, QwenOFT, $\pi_{0.5}$) policies with SFT and online actor-critic RL with human intervention.
- Improved real-robot millimeter-level task success from **10% to 85%** through iterative hardware RL.

Research Assistant, Georgia Institute of Technology **Atlanta, GA**
Inverse Reinforcement Learning for Reusable Motor Skills, supervised by Dr. Anqi Wu *Aug 2024–Jun 2025*

- Developed an inverse reinforcement learning framework that decomposes behavior into reusable motor motifs and recovers reward structure from trajectory data.
- Applied EM to multi-reward MDPs to infer latent behavioral intentions from continuous animal trajectories.

Research Assistant, Tsinghua University & New York University **Beijing, China & New York, NY**
Reinforcement Learning Models of Replay, supervised by Dr. Sen Song & Dr. Marcelo Mattar *May 2023–Jul 2025*

- Built a modular reinforcement learning model in **JAX** that reproduces replay and its roles in planning and cognitive-map learning.
- Analyzed learned neural manifolds and reproduced biological replay experiments to connect model dynamics with experimental observations.

Research Assistant, Shanghai Qizhi Institute **Beijing, China**
E/I-Separated Recurrent Neural Networks, supervised by Dr. Lu Mi *Jul 2025–May 2026*

- Investigate how Dale constraints can affect learning in recurrent networks.

SELECTED AWARDS:

National Scholarship (2/165), Samsung Scholarship (20/160), National Chemistry Olympiad Silver Medal.

SELECTED PROJECTS:

Robotics: Stochastic Residual Dynamics for Latent World Models: Flow-Matched Residual Kernels on a Frozen LeWorldModel

CV: Visual straightening in AI and brain.

RL: Honor of Kings (MOBA game, both Single- and Multi-agent) with PPO, COMA, QMIX. Supply chain management with VDN. Mahjong Competition with PPO and SFT.